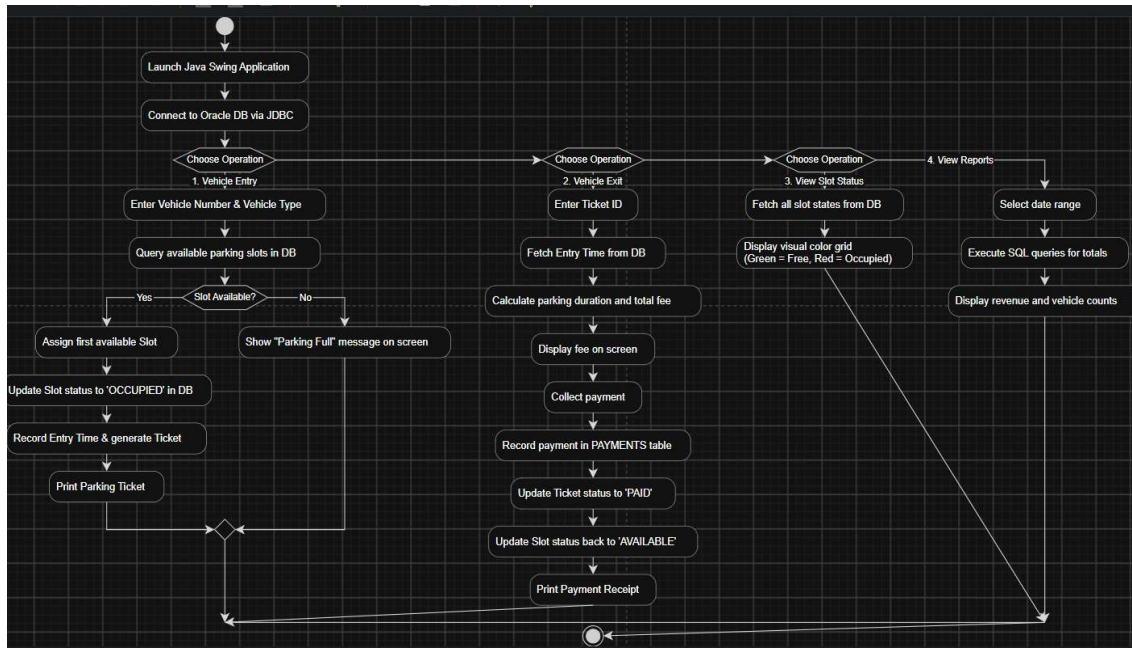
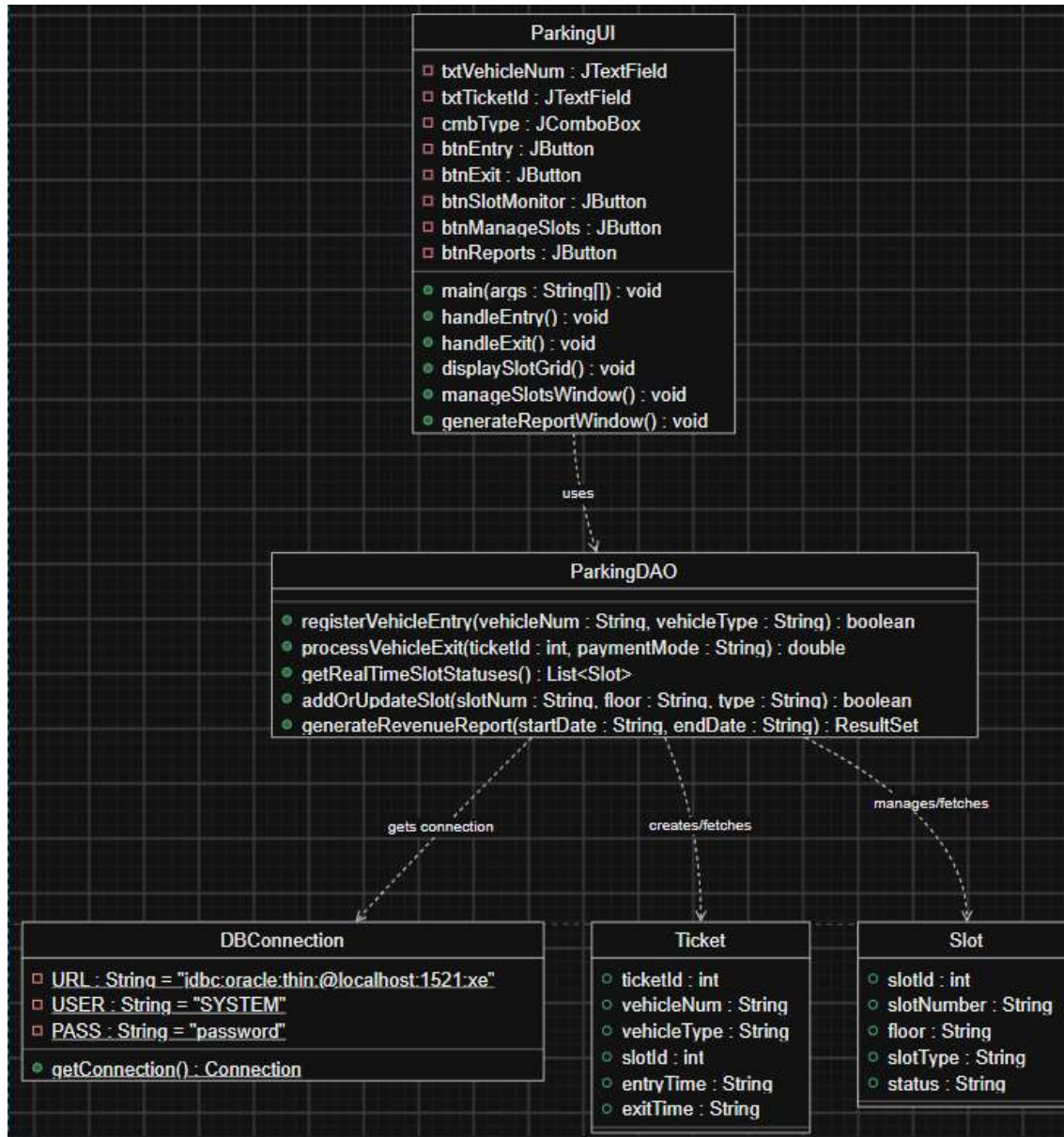


JAVA MINI PROJECT

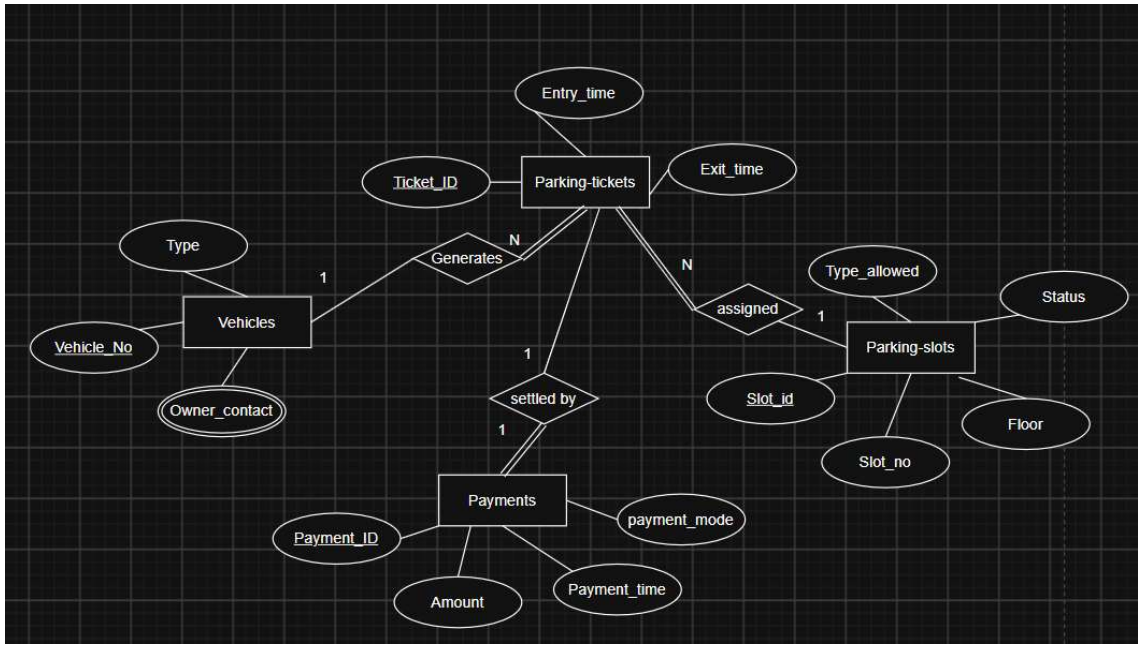
Activity diagram:



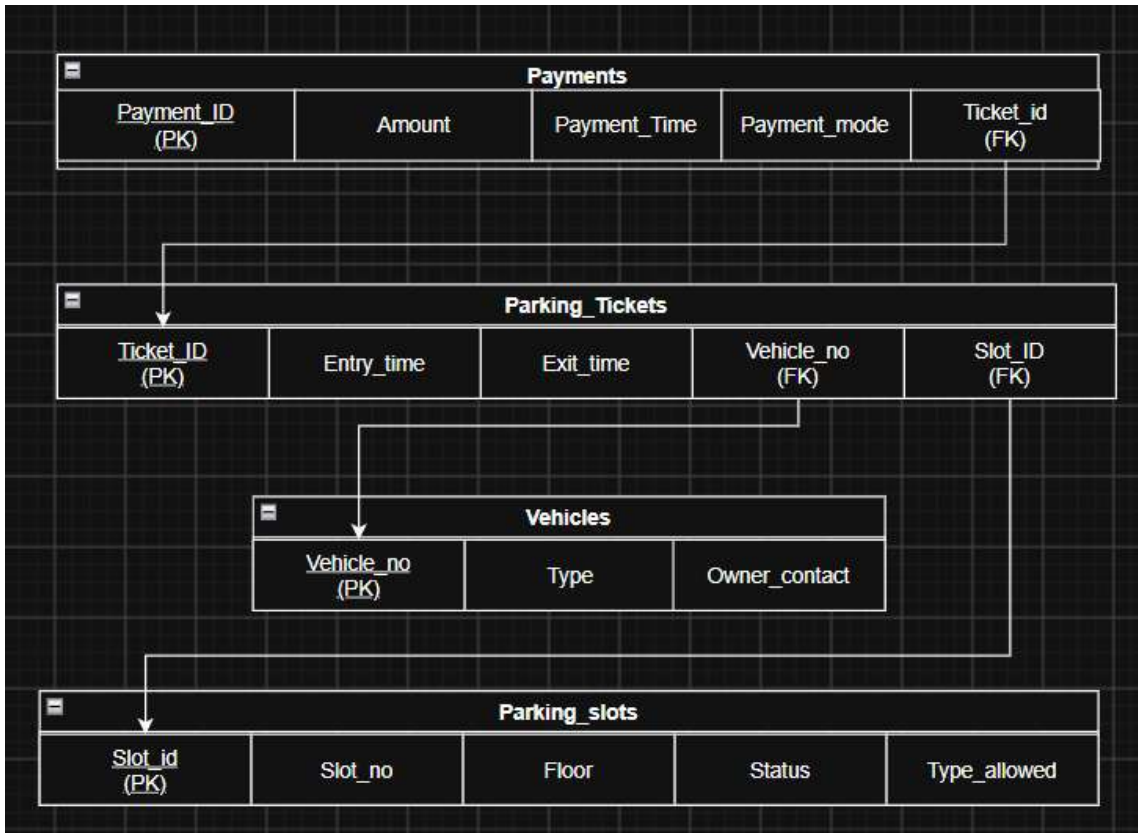
Class diagram:



ER Diagram:



Schema diagram:



Functional dependencies:

Vehicles

FD_1 : Vehicle_No $\rightarrow$ Type, Owner_contact

Parking_slots

FD_1 : Slot_id $\rightarrow$ Slot_no, Floor, Status, Type_allowed

FD_2 : {Slot_no, Floor} $\rightarrow$ Slot_id, Status, Type_allowed

Parking_tickets

FD_1 : Ticket_ID $\rightarrow$ Entry_time, Exit_time, Vehicle_No, Slot_id

Payment

FD_1 : Payment_ID $\rightarrow$ Amount, Payment_time, payment_mode, Ticket_ID

FD_2 : Ticket_ID $\rightarrow$ Payment_ID, Amount, Payment_time, payment_mode

1. Primary Actors

- **Parking Attendant / Operator:** Manages daily vehicle entries, exits, and payment collection at the gate.
- **System Administrator:** Oversees slot configurations, user access, and system reports.
- **System (Automated Processes):** Handles real-time slot state calculations, duration tracking, and fee computations via JDBC/Oracle.

2. Core Use Cases

UC-01: Vehicle Entry & Parking Assignment

- **Primary Actor:** Parking Attendant
- **Description:** Records an incoming vehicle and assigns an available parking slot.
- **Preconditions:** The Swing UI is running, and the Oracle database connection is active.
- **Main Success Scenario:**
 - Driver approaches the entrance.
 - Attendant enters vehicle details (Vehicle Registration Number, Vehicle Type e.g., 2-Wheeler / 4-Wheeler) into the Java Swing form.
 - System queries Oracle DB (JDBC) to fetch available slots matching the vehicle type.
 - System assigns the first available slot and updates its status to OCCUPIED.
 - System records the entry record (Vehicle_ID, Slot_ID, Entry_Time) in the database.
 - System generates and prints a parking ticket/receipt with a unique Ticket ID and entry timestamp.

- **Extensions / Alternate Flow:**
 - **3a. No slots available:** System displays a "Parking Full" warning on the Swing UI. Attendant informs the driver, and entry is aborted.

UC-02: Vehicle Exit & Payment Processing

- **Primary Actor:** Parking Attendant
- **Description:** Processes a departing vehicle, calculates parking fees based on duration, and frees the parking slot.
- **Preconditions:** Vehicle entry record exists in the database with status **PARKED**.
- **Main Success Scenario:**
 1. Driver approaches the exit gate and presents the parking ticket.
 2. Attendant enters the Ticket ID or Vehicle Number into the Swing UI.
 3. System retrieves entry details from Oracle DB and captures the current time (**Exit_Time**).
 4. System calculates total duration and computes the fee according to defined hourly rates.
 5. Fee amount is displayed on the Swing UI.
 6. Attendant collects payment (Cash/Card) and selects the payment mode on the UI.
 7. System logs the transaction details (**Payment_ID**, **Amount**, **Payment_Mode**, **Timestamp**) into Oracle DB.
 8. System updates the slot status back to **AVAILABLE** and marks the vehicle status as **EXITED**.
 9. System prints a payment receipt for the driver.

UC-03: Real-Time Slot Availability Monitoring

- **Primary Actor:** Parking Attendant / Administrator
- **Description:** Provides a visual dashboard of all parking slots and their current occupancy status.
- **Preconditions:** UI dashboard panel is initialized.
- **Main Success Scenario:**
 1. Attendant views the Swing dashboard panel showing a visual grid/map of parking slots.
 2. System fetches real-time slot states (**AVAILABLE**, **OCCUPIED**, **UNDER_MAINTENANCE**) from Oracle DB via JDBC.
 3. UI updates slot panels dynamically (e.g., Green for Available, Red for Occupied).
 4. User can click on a slot to view occupying vehicle details and time elapsed.

UC-04: Manage Parking Slot Configurations

- **Primary Actor:** System Administrator
- **Description:** Allows administrators to add, modify, or disable parking slots within the system.
- **Preconditions:** Admin is logged into the system with administrative privileges.

- **Main Success Scenario:**
 1. Admin opens the Slot Management window in the Java Swing UI.
 2. Admin enters new slot details (Slot Number, Floor/Zone, Vehicle Type Supported).
 3. Admin clicks "Save".
 4. System executes SQL insert/update commands via JDBC to store changes in Oracle DB.
 5. Swing UI reflects updated slot count and structure.

UC-05: Revenue & Occupancy Report Generation

- **Primary Actor:** System Administrator
- **Description:** Generates analytical summaries of parking usage and total revenue collected over a selected timeframe.
- **Preconditions:** Past entry and payment records exist in Oracle DB.
- **Main Success Scenario:**
 1. Admin selects a date range (Daily, Weekly, Monthly) on the Reporting UI.
 2. System executes aggregate SQL queries (**SUM**, **COUNT**, **GROUP BY**) against payment and entry tables.
 3. System formats and displays total revenue, total vehicles served, peak occupancy hours, and breakdown by vehicle type on the screen.
 4. Admin can export or print the report.

